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Optimizing Sustainable Tourism: A Multi-Objective Framework for Juneau and Beyond

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Abstract

This study develops a multi-dimensional sustainable tourism optimization framework for Juneau, Alaska, integrating economic, social, and environmental dimensions to balance tourism-driven prosperity with ecological and socio-cultural integrity. Utilizing a hybrid Analytic Hierarchy Process and entropy weighting method, the model assigns robust indicator weights. Optimized via the NSGA-II algorithm, it identifies an optimal tourist threshold, achieved through a strategic tax adjustment. This policy not only sustains economic revenue at USD 325 million but also funds a critical feedback loop: revenue reinvestment into environmental conservation and social infrastructure, which stabilizes cost indices and enhances community well-being. The model's projections show this approach significantly mitigates environmental degradation, notably glacier retreat, and alleviates social pressures such as infrastructure overload and resident dissatisfaction. A key contribution of this research is the framework's adaptability, which was validated through its application to Barcelona, Spain. There, the framework was recalibrated with social indicators tailored to address urban overtourism, achieving substantial reductions in housing and congestion costs alongside environmental improvements, while economic recovery was maintained. Sensitivity analyses confirm the model's stability, though data limitations and subjective weighting suggest future enhancements via real-time analytics and dynamic modeling. Key policy recommendations include dynamic tourist caps, diversified attractions, and community engagement platforms, offering scalable solutions for global tourism destinations. This framework advances sustainable tourism by providing a blueprint to decouple economic growth from ecological and social harm, ensuring the longevity of natural and cultural assets amidst climate challenges.

Keywords: sustainable tourism; multi-objective optimization; economic–environmental–social balance; AHP–entropy weighting; tourist carrying capacity

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Academic Editors: Harry Coccossis and Jun (Justin) Li

Received: 26 June 2025
Revised: 30 July 2025
Accepted: 6 August 2025
Published: 14 August 2025

Citation: Pan, J.; Yang, H.; Wang, Z.; Peng, B.; Li, S. Optimizing Sustainable Tourism: A Multi-Objective Framework for Juneau and Beyond. *Sustainability* **2025**, *17*, 7344. https://doi.org/10.3390/su17167344

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1. Introduction

Sustainable development has emerged as a cornerstone of modern policy and research since the United Nations introduced the Sustainable Development Goals (SDGs)

in 2015 [1]. Within this framework, sustainable tourism stands out as a critical domain, tasked with balancing economic prosperity, ecological preservation, and socio-cultural integrity. This delicate balance is not a static endpoint but a dynamic process of adaptation, a perspective that resonates deeply with the challenges faced by tourism-dependent regions worldwide [2]. Among these, Juneau, the capital of Alaska, exemplifies the dual-edged nature of tourism growth. The city's breathtaking natural landscapes, including the iconic Mendenhall Glacier, draw millions of visitors annually, injecting significant revenue into the local economy. Yet, this very influx precipitates acute environmental degradation and social strain, necessitating a rigorous, science-driven approach to sustainable tourism planning [3].

Juneau's tourism boom, particularly in the post-COVID-19 era, underscores both its opportunities and vulnerabilities [4]. In recent years, the city has seen over 700 cruise ship arrivals annually, ferrying nearly 1.7 million passengers and crew to its docks. This surge has bolstered Juneau's economy, generating approximately 4000 jobs and USD 200 million in labor income [5,6]. However, the environmental toll is stark. The Mendenhall Glacier, a flagship attraction, has retreated by an area equivalent to eight football fields since 2007, a decline accelerated by rising local temperatures linked to excessive tourism and associated carbon emissions [7]. Beyond glacial retreat, the industry's ecological footprint manifests in diminished air quality and resource depletion [8]. Socially, the seasonal influx strains public infrastructure and impacts residents' quality of life, raising critical questions about the long-term viability of unchecked tourism growth [9].

This tension between economic gain and socio-environmental cost is a global dilemma, with the tourism sector's GDP contribution expected to reach nearly USD 11 trillion by 2026 [10]. In response, academic research has long sought to define and manage tourism's impacts, beginning with the foundational concept of Tourism Carrying Capacity (TCC), which aims to establish maximum visitor levels without causing irreversible harm [11]. Building on this, scholars have developed more sophisticated, multi-dimensional frameworks [12,13]. Methodologies such as the driving force–pressure–state–impact–response (DPSIR) model have been combined with tripartite social–economic–environmental lenses to create comprehensive indicator systems [14]. Similarly, assessment models integrating economic, ecological, and political factors and benchmarking tools like the STBT have advanced our ability to evaluate sustainability [15,16]. However, despite these advances, three critical gaps remain. First, most existing models excel at static assessment but lack the capacity for dynamic optimization; they can diagnose a problem but cannot prescribe an optimal path forward. Second, many frameworks are designed to be universal, often failing to offer the customization and adaptability needed to address the unique pressures of specific locales, such as the cruise-dominated port environment of Juneau [17]. Third, effectively modeling and navigating the trade-offs between conflicting objectives—such as maximizing revenue while minimizing glacial melt—remain significant challenges.

To address these gaps, this study develops and validates a dynamic, multi-objective optimization framework for sustainable tourism management. Building on prior calls for integrated systems [18], our framework advances the field in three key ways. First, it employs a hybrid Analytic Hierarchy Process (AHP) and entropy weighting method to capture the specific priorities and complexities of a destination, enhancing adaptability. Second, it utilizes the NSGA-II algorithm to move beyond static assessment, identifying a range of Pareto-optimal solutions that navigate the trade-offs between conflicting economic, social, and environmental objectives. Third, it provides a robust, science-driven tool for proactive policymaking. We first apply this framework to the urgent case of Juneau and then test its adaptability by recalibrating and applying it to Barcelona, a destination

with markedly different sustainability challenges. Therefore, the specific objectives of this study are as follows: (1) to quantify the non-linear effects of tourist volume on the local environment, exemplified by glacier retreat in Juneau; (2) to identify an optimal tourist threshold that maintains equilibrium across the economic, social, and environmental dimensions; and (3) to test the framework's adaptability and robustness by applying it to cities with distinct characteristics.

2. Method

2.1. Data Collection and Preparation

2.1.1. Research Hypotheses

H1. *Linearity of Carbon Emissions: A linear relationship is assumed between tourist volume and local tourism-related carbon emissions for model tractability.*

H2. *Isolation of Local Tourism Impact on Glacier Melt: To isolate the marginal impact of local tourism, the rate of glacier retreat is modeled as a function of tourist activity, with the broader effects of global climate change treated as a constant baseline.*

H3. *Logistic Function for Social Pressure: A logistic function is used to model the non-linear relationship between infrastructure pressure and resident satisfaction, capturing the effects of system saturation.*

H4. *Homogeneity and Policy-Dependence of Tourist Spending: Tourist spending is assumed to be homogeneous and is modeled as a function of the local tax rate, which serves as the primary economic policy lever within this study.*

2.1.2. Data Collection

During the data collection and preprocessing phase, we gathered key indicators for Juneau, Alaska, and Barcelona, Spain, over the past five years (2019–2024), including tourist numbers, economic revenue, air quality, and other relevant metrics. These data were sourced from official survey reports, municipal databases, and regional tourism statistics. To address missing values for certain critical indicators, we employed time series models and regression analysis to estimate and impute the incomplete data, ensuring dataset completeness and reliability for subsequent analysis. The specific indicators, their corresponding sustainability dimensions, and their precise sources are detailed in Table 1.

Table 1. Data resources of this study.

Dimension	Indicator	Location	Source/Agency	Reference/URL
Economic	Annual Tourism Revenue	Juneau	City and Borough of Juneau, Finance Department	https://juneau.org/finance (accessed on 15 March 2025)
	Tourism-Related Employment	Juneau	Alaska Department of Labor and Workforce Development	https://labor.alaska.gov/ (accessed on 15 March 2025)
	Annual Tourist Arrivals	Juneau	City and Borough of Juneau, Docks & Harbors	https://juneau.org/harbors (accessed on 15 March 2025)
	Tourism Tax Revenue	Barcelona	Barcelona City Council (Ajuntament de Barcelona)	https://ajuntament.barcelona.cat/ (accessed on 15 March 2025)

Table 1. *Cont.*

Dimension	Indicator	Location	Source/Agency	Reference/URL
Social	Resident Satisfaction Index	Juneau	Supplemental Survey Data/Literature	(Specify if from a specific study or survey)
	Infrastructure Pressure (e.g., traffic)	Juneau	City and Borough of Juneau, Community Development	https://juneau.org/community-development (accessed on 15 March 2025)
	Housing Price Index	Barcelona	Spanish National Statistics Institute (INE)	https://www.ine.es/ (accessed on 15 March 2025)
	Population Density	Barcelona	Statistical Institute of Catalonia (Idescat)	https://www.idescat.cat/ (accessed on 15 March 2025)
Environmental	Mendenhall Glacier Retreat Rate	Juneau	NOAA, National Centers for Environmental Information	https://www.ncdc.noaa.gov/cdo-web/ (accessed on 15 March 2025)
	Air Quality Index (AQI)	Juneau	U.S. Environmental Protection Agency (EPA)	https://www.airnow.gov/ (accessed on 15 March 2025)
	Air Quality Index (AQI)	Barcelona	European Environment Agency (EEA)/Local Station	https://www.eea.europa.eu/ (accessed on 15 March 2025)
	Carbon Emissions (Tourism Sector)	Both	Estimated Based on Transport/Energy Data from Sources Above	(Model-derived)

2.1.3. Data Preparation

To ensure consistency across diverse data with varying scales and magnitudes, we applied normalization techniques to prevent any single feature from disproportionately influencing the model outcomes. The normalization process involved two steps: positive orientation and standardization.

Positive Orientation

(1) The primary purpose of data transformation is to convert inverse indicators (where smaller values are better) into positive indicators (where larger values are better). This unifies the direction of all indicators to facilitate comprehensive evaluation and model optimization.

(2) For non-positive indicators, such as infrastructure pressure, living costs, and carbon footprint, we applied a transformation formula:

$$x_i = \max\{a_1, a_2 \ldots, a_n\} - a_i, i = 1, 2 \ldots, n \quad (1)$$

where a_i is the value of the (n)-th indicator for the (i)-th evaluation object, and $\max(a_i)$ is the maximum value of the (n)-th indicator across all objects.

(3) For intermediate indicators, such as tourist numbers, population size, and tourism employment, which have optimal ranges, we used

$$M = \max\{a - \min\{x_i\}, \max\{x_i\} - b\}, \tilde{x_i} = \begin{cases} 1 - \frac{a - x_i}{M} & , x_i < a \\ 1 & , a \le x_i \le b \\ 1 - \frac{x_i - b}{M} & , x_i > b \end{cases} \quad (2)$$

where a and b are, respectively, the upper and lower bounds of the optimization interval, and M represents the maximum deviation value within an indicator.

Standardization

The primary purpose of data standardization is to eliminate dimensional differences among the various features, enabling them to be compared and analyzed on a common scale.

After positive orientation, the data were organized into a matrix. The standardized matrix (X) was computed with the following elements: $X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}$.

If the normalized matrix is Z, then the elements in Z are $Z_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^{n} x_{ij}^2}}$.

The data used in the environmental and social cost models are normalized in the following modeling process.

2.2. Establishment of the Evaluation Framework

2.2.1. Weight Determination

Analytic Hierarchy Process

The Analytic Hierarchy Process (AHP) is a multi-criteria decision-making method designed to address complex decision problems [19]. It structures the problem into a hierarchy, with the top level as the goal—selecting the optimal strategy for achieving sustainable tourism development in Juneau. The bottom level comprises the scheme layer, encompassing 11 influencing factors, while the intermediate level includes the criteria layer, consisting of three dimensions: economic, social, and environmental. By integrating quantitative and qualitative approaches, the AHP constructs pairwise comparison matrices, performs consistency checks, and derives weights to achieve the hierarchical objectives. The evaluation framework for Juneau's sustainable tourism is illustrated in Figure 1.

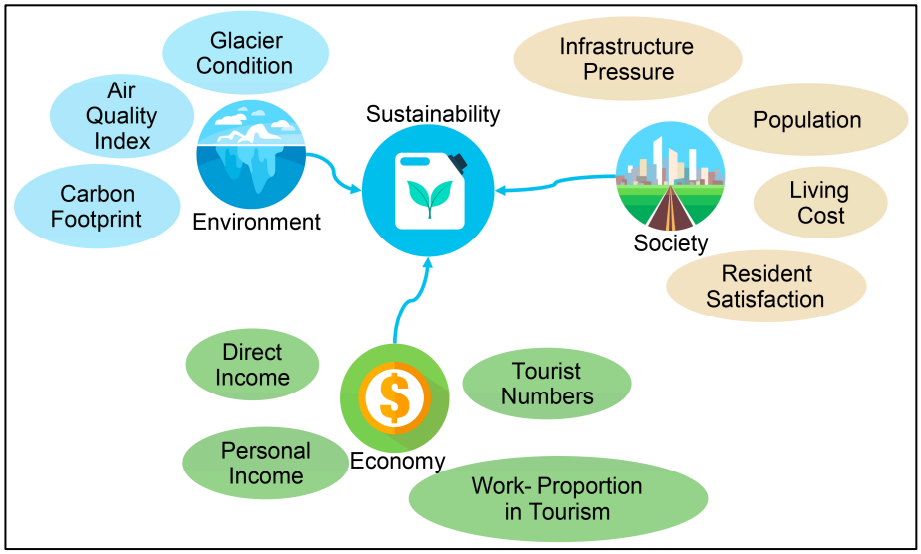


Figure 1. Evaluation framework for sustainable tourism in Juneau.

The AHP method involves three main steps:

(1) Constructing the Judgment Matrix:

The matrix is defined as

$$A = (a_{ij})_{n\times n} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad (3)$$

where $a_{ij} \times a_{ji} = 1$, and a_{ij} denotes the relative importance of indicator (i) over (j).

To ensure consistent scoring across Juneau's diverse indicators, the evaluation data are normalized using the min–max method:

$$r_i = \frac{a_i - a_{\min}}{a_{\max} - a_{\min}} \quad (4)$$

where r_i is the normalized result, a_i is the evaluation data, a_{max} is the maximum value of the evaluation data, and a_{min} is the minimum value.

(2) Calculating Weights:

The eigenvector, ω, and the maximum eigenvalue, λ_{max}, of the judgment matrix were determined. Through subjective scoring of indicators, combined with normalization and the 1–9 scale method, the corresponding judgment matrix was constructed. The calculations yielded

Eigenvector: $\omega = [1.302,\ 0.79,\ 0.908]^{\mathrm{T}}$;

Weights for the criteria layer indicators: [0.43411, 0.26323, 0.30266];

Maximum eigenvalue: $\lambda_{max} = 3.022$.

(3) Consistency Check:

For matrices with dimensions greater than 2, a consistency check is required, using the following formulas:

$$CI = \frac{\lambda_{\max} - n}{n-1}, CI = \begin{cases} 0, complete\ consistency \\ approaching\ 0, Excellent\ consistency \\ Bigger, worse\ consistency \end{cases} \quad (5)$$

To evaluate CI, the random consistency index (RI) is introduced:

$$RI = \frac{\frac{\sum_{i=1}^{m} \lambda_i}{m} - n}{n-1} \quad (6)$$

The consistency ratio, $CR = \frac{CI}{RI}$, is defined if $CR < 0.1$, and the matrix is considered acceptably consistent; otherwise, it must be revised and recalculated.

Entropy Weighting Method

Although the Analytic Hierarchy Process (AHP) is a well-established method for determining indicator weights, it relies heavily on subjective judgment, lacking objectivity [20]. To achieve more accurate weighting, we adopted a hybrid subjective–objective approach for Juneau's sustainable tourism evaluation system, using the entropy weighting method to assign weights to second-level scheme-layer indicators.

Originating from thermodynamics and introduced to information theory by Shannon, the entropy weighting method assesses the variability of indicators, calculating entropy-based weights to derive relatively objective indicator weights.

Given (n) evaluation objects and (m) indicator variables, the value of the (j)-th indicator for the (i)-th object is $a_{ij}(i = 1, 2 \ldots, n; j = 1, 2 \ldots, m)$, forming the data matrix $\mathrm{A} = \left(\mathrm{a_{i\times j}}\right)_{\mathrm{n\times m}}$.

The entropy weighting method involves the following steps:

(1) Calculate $p_{ij}(i = 1, 2 \ldots, n; j = 1, 2 \ldots, m)$, the proportion of the (i)-th object for the (j)-th indicator, using the original data matrix:

$$p_{ij} = \frac{a_{ij}}{\sum_{i=1}^{n} a_{ij}}, i = 1, 2 \ldots, n; j = 1, 2, \ldots, m \tag{7}$$

(2) Compute the entropy value of the (j)-th indicator:

$$e_j = -\frac{1}{\ln(n)} \sum_{i=1}^{n} p_{ij} \cdot \ln(p_{ij}), j = 1, 2, \ldots, m \tag{8}$$

(3) Determine the information utility value of the (j)-th indicator:

$$d_j = 1 - e_j, j = 1, 2, \ldots m \tag{9}$$

(4) Calculate the weight of the (j)-th indicator:

$$\omega_j = \frac{g_{\mathrm{j}}}{\sum_{j=1}^{m} g_i}, j = 1, 2, \ldots, m \tag{10}$$

2.2.2. Economic, Social, and Environmental Dimensions

To apply this framework, we selected 11 specific indicators tailored to Juneau's unique context (Figure 1). The principles guiding this selection included scientific relevance, systems thinking, and data accessibility. The chosen indicators and the rationale for their inclusion are detailed in the Supplementary Materials.

The model's core concept is a constrained optimization problem designed to maximize economic benefits. This is achieved by simultaneously requiring that key environmental and social indicators be maintained above predefined sustainability thresholds.

(1) Economic Model

The economic dimension focuses on tourism revenue, driven by per capita tourist spending and visitor numbers, offset by city service costs. The net revenue increase ($R(x, r)$) is modeled as

$$R(x, r) = M \times x \times r \tag{11}$$

where $R(x, r)$ denotes the total revenue available for the city inputs under certain conditions; the number of tourists is x; the city tax rate is r; and x represents the number of tourists, which can be expressed as:

$$x = x_{max} \cdot (1 - a \times r - b \times A - c \times E) \tag{12}$$

where x_{max} is the maximum number of travelers, A is the social situation, and E shows the environmental situation.

(2) Social Model

Residents are central to urban development, and excessive social costs reduce satisfaction, hindering sustainability. We quantified social costs using

$$S(x) = \frac{K}{1 + e^{-r \cdot (x - x_0)}} + a \cdot (x - \lambda)^n + \epsilon + k_1 \times \alpha \times R \tag{13}$$

where K is the maximum number of tourists per year, λ is an undefined parameter, ε is a perturbation term, and the last term is the feedback of additional income on social evaluation. Under the constraints of the existing resource conditions, the social cost is positively related to the number of people, which can be fitted by a logistic regression model; the second term of the equation indicates that the social cost of overtourism grows in a power function trend.

(3) Environmental Model

Environmental quality is a key factor affecting the lives of urban residents and the long-term attraction of natural scenery to tourists, and it is the basis for sustainable development. In evaluating the environmental conditions, we selected three main factors, namely, air quality index, glacier melting rate, and carbon footprint, and used the following equations to quantitatively evaluate them:

$$E(x) = d \times L(x) + f \times x^n + \epsilon + k_2 \times (1 - \alpha) \times R \tag{14}$$

where d and f are undefined parameters, ε is a perturbation term, and the last term is the feedback of additional income on social evaluation.

A search shows that the glacier melting rate, $L(x)$, can be approximated as

$$L(x) = d \times (T_0 + C \times x) \times \mathrm{e}^{x2} \tag{15}$$

where d is the glacier sensitivity factor, T_0 is the initial temperature, and C is the annual per capita carbon emissions.

Carbon emissions are calculated from city-related gas emissions data and can be expressed as follows:

$$\mathrm{C} = (CO_2 \times 1 + CH_4 \times 28 + \mathrm{N_2O} \times 256)/\mathrm{kg} \tag{16}$$

Based on data from the City of Juneau's gas emissions profile, it can be concluded that the average carbon emissions are 66.13 kg per person per day.

(4) Constraints

Tourist number constraints:

In order to prevent the city from being subjected to overtourism with irreversible social or environmental consequences, we limit the maximum number of tourists per year, K, and K satisfies

$$x(t) = \frac{K}{1 + (\frac{K}{N_0} - 1) \cdot e^{-rt}} \tag{17}$$

$K < 1.8$ million.

Economic Constraints:

To mitigate the escalating growth in tourist numbers and ensure sustainable tourism development while supporting continuous investments in local infrastructure and environmental improvements, the government implements policies such as adjusting tax rates and increasing tourist expenditures. We assume that the city influences tourist numbers (x) by raising the tax rate (r), where $r > 0$.

2.2.3. Objective Function

The model seeks to maximize economic revenue while minimizing adverse environmental impacts, alleviating economic pressures, and reducing social burdens. To achieve this, the objective function is formulated as a multi-objective optimization problem, balancing revenue against environmental costs, infrastructure expenses, and social costs to

maximize overall benefits. This function, termed the objective function, encapsulates trade-offs and optimizations across all dimensions. Accordingly, the objective function is defined as

$$minZ = -\omega_1 \times \frac{R}{R_{\max}} + \omega_2 \times \frac{E}{E_{\max}} + \omega_3 \times \frac{S}{S_{\max}}$$
$$s.t. \begin{cases} r > 0 \\ 0 < x < K \\ R_{\min} < R \\ L < L_{\max} \\ I < I_{\max} \\ S < S_{\max} \end{cases} \quad (18)$$

where w_i denotes the importance of the economy, the environment, and the environment to the goal of optimization; R denotes annual government revenue; E denotes environmental costs; and S denotes social costs.

To determine the weights, W_i, for the economic, social, and environmental scores in the objective function, we evaluated several methods, including the Analytic Hierarchy Process (AHP), Principal Component Analysis, entropy weighting, and standard deviation approaches. These methods are categorized into subjective approaches, which rely on expert judgment and are well established, and objective approaches, which are data-driven but less refined. To enhance accuracy, we adopted a hybrid approach for Juneau's sustainable tourism evaluation system, combining subjective and objective techniques. Specifically, the AHP was used to assign weights to first-level indicators (economic, social, and environmental dimensions), while the entropy weighting method determined weights for second-level sub-indicators, ensuring a robust and balanced weighting scheme.

2.3. Multi-Objective Optimization

The sustainable tourism evaluation model was formulated as a multi-objective optimization problem to balance economic revenue, environmental costs, and social costs. We combined the weighted sum method with Pareto optimization to identify optimal solutions, employing sensitivity analyses to assess the impact of key parameters on model outcomes:

$$min\ Z = -\omega_1 \times \frac{R}{R_{\max}} + \omega_2 \times \frac{E}{E_{\max}} + \omega_3 \times \frac{S}{S_{\max}} \quad (19)$$

The development of scenic areas and their host cities results from interactions between multiple factors, with tourist numbers (x) being a critical driver of environmental pollution, revenue growth, and resident pressure. Thus, we selected x as the independent variable to generate fitted functions for each dimension's evaluation model. To explore trade-offs between objectives, we applied the Pareto optimality approach. The Pareto frontier represents a set of non-dominated solutions, where improving one objective (e.g., revenue) worsens another (e.g., social costs). Using the NSGA-II algorithm, we generated the Pareto frontier, identifying optimal trade-offs between economic revenue, environmental costs, and social costs. The resulting GE matrix is shown in Figure 2.

In the coordinate system, S represents the social score, R denotes the revenue score, and E indicates the environmental score. Red points signify optimal solutions where all dimensions achieve ideal conditions during the optimization process, while green points represent practical solutions attainable through policy adjustments and revenue feedback, forming the Pareto frontier solution set.

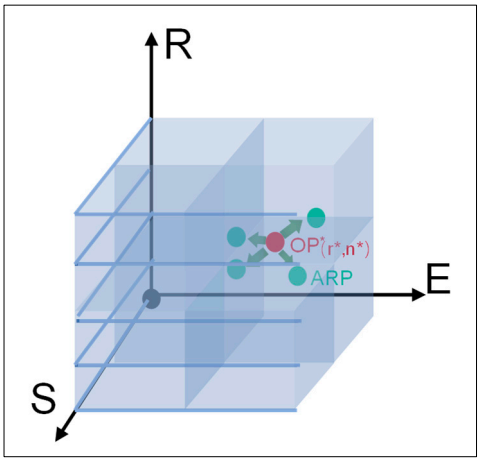


Figure 2. Three-dimensional analysis cube for the evaluation model. In OP(r*, n*), the asterisk (*) denotes the optimal value or optimal solution.

2.4. Assumptions and Notations

Tourism systems involve dynamic interactions between factors like climate change, tourist behavior, and policy interventions [21,22]. To derive practical conclusions within a limited research scope, we simplified the system using reasonable assumptions and established a clear notation system to support mathematical modeling.

Assumption 1. *Cumulative Revenue Effects.*

Increased tax revenue (e.g., for environmental or infrastructure investments) has a delayed impact on environmental and social indicators, with a negligible short term (1–2 years) but significant long term, due to ecological and governance lags.

Assumption 2. *Infrastructure Investment Benefits.*

Infrastructure investments (e.g., public transport and renewable facilities) enhance Juneau's appeal and reduce per capita carbon emissions (from 66.13 to ~50 kg/person/day), attracting eco-conscious tourists.

Assumption 3. *Proportional Emissions.*

Carbon emissions scale linearly with tourist numbers (66.13 kg/person/day) within a threshold, below which environmental damage is reversible, aligning with carrying capacity principles.

Table 2 defines key symbols (e.g., x: tourist numbers; R: revenue), ensuring clarity across economic, social, and environmental models.

Table 2. Nomenclature employed in model development.

Symbol	Description	Unit
x	Annual Tourist Numbers	USD 10,000/year
M	Average Tourist Expenditure	
R	Annual Government Revenue	USD 10,000
r	Tax Rate Increase for Related Industries	%
L	Glacier Melting Rate	Tons/year
E	Environmental Cost	Dimensionless
C	Carbon Emissions	kg/person × day
S	Social Cost	Dimensionless

2.5. Model Application to Barcelona

To validate the adaptability and scalability of the framework developed for Juneau, we deliberately selected Barcelona, Spain, as a contrasting case study. Barcelona represents a classic example of "overtourism" in a dense, urban–cultural heritage destination [23,24]. Unlike Juneau, where the primary sustainability challenge is environmental, Barcelona's critical issues are predominantly socio-economic, driven by intense tourist inflows that have been widely documented to strain public services and erode resident quality of life [25].

This fundamental difference provides a rigorous test for our model. The core hypothesis is that our framework is not a fixed solution but a flexible tool that can be recalibrated for diverse destination typologies. Therefore, while the multi-objective optimization structure was retained, the indicator system required significant modification to align with Barcelona's local context [26,27]. Based on documented challenges, such as housing market distortion and traffic congestion, our recalibrated model prioritizes social indicators as the primary influencing factors, as detailed in the subsequent analysis:

(1) Economic Model

Economic income exhibits consistency, and the previous formula is retained in Equation (11). The number of tourists, x, is given by Equation (12).

(2) Environmental Cost Model

$$E(x) = f \cdot e^{x-f(x)} + \epsilon + k_1 \times \alpha \times R \tag{20}$$

Here, f represents the environmental penalty coefficient under local conditions; $f(x)$ is a monotonically decreasing function of the number of tourists, influenced by the self-recovery capacity of the local ecological environment; and the final term accounts for the feedback effect of additional income on environmental costs.

(3) Social Cost Model

$$S(x) = j \times x + k \times P + m \times N + k_2 \times (1 - \alpha) \times R \tag{21}$$

Here, j, k, and m represent the weight coefficients obtained through the entropy weight method; x denotes the annual number of tourists; P signifies the local average housing price; N represents the noise coefficient; and the final term accounts for the feedback effect of additional income on social costs.

(4) Target Variable

To solve the multi-objective optimization of this model, we continue to employ the weighted sum method (Equation 19).

3. Results

3.1. Model Validation and Composite Weight Results

Composite weights for the model's indicators were computed using a hybrid approach that combined the Analytic Hierarchy Process (AHP) for primary dimensions with the entropy weight method for secondary indicators. This process yielded primary weights of 0.43411 for the economic dimension, 0.30266 for the environmental dimension, and 0.26323 for the social dimension. The reliability of the AHP component was validated through a consistency test, which resulted in a consistency ratio (CR) of 0.021, confirming that the judgment matrix met the required consistency threshold. A comprehensive summary of the weight assignments for all primary and secondary indicators is presented in Table 3.

Table 3. Results of AHP consistency test.

Consistency Test Results			
CI	**RI**	**CR**	**Result**
0.011	0.525	0.021	Pass

The composite weights derived from the hybrid AHP–entropy method, detailed in Table 4, reflect the relative importance of each secondary indicator within the overall evaluation framework. The results highlight several key priorities.

Table 4. Composite weight calculations.

Primary Indicator	Primary Weight	Secondary Indicator	Secondary Weight	Composite Weight
Economic	0.43411	Direct Revenue	0.22879	0.0993
		Personal Income	0.24824	0.1078
		Tourism Employment Proportion	0.29450	0.1278
		Tourist Numbers	0.22845	0.0992
Social	0.26323	Population Size	0.15325	0.0342
		Infrastructure Pressure	0.31346	0.0825
		Living Cost	0.25181	0.0562
		Resident Satisfaction	0.27148	0.0714
Environmental	0.30266	Air Quality Index	0.45399	0.1374
		Glacier Condition	0.31049	0.0940
		Carbon Footprint	0.23552	0.0713

Among all indicators, the air quality index exhibited the highest composite weight (0.1374), followed closely by the proportion of tourism-related employment (0.1278). This underscores that balancing environmental health and local economic opportunity is the model's top priority for Juneau. Conversely, population size had the lowest composite weight (0.0342), suggesting that it is a less critical factor in this specific context. At the dimensional level, the economic indicators collectively contributed the largest share to the overall weights, while the social dimension's indicators ranked the lowest.

3.2. Evaluation Models for Economic, Social, and Environmental Dimensions

The sustainable tourism evaluation model was calibrated to quantify the relationships between tourist numbers (x) and the scores of the economic, social, and environmental dimensions for Juneau. Optimized solutions were derived using the NSGA-II algorithm, with key trends presented in Figures 3–5.

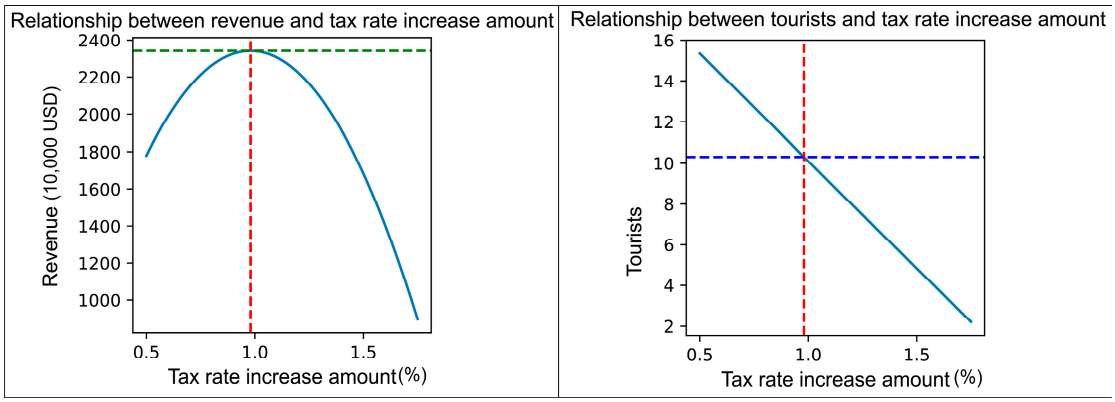


Figure 3. Relationships between dimensional indicators and tourist numbers.(The intersection of the red dashed line and the green dashed line represents a potential optimal trade-off point on the Pareto front.)

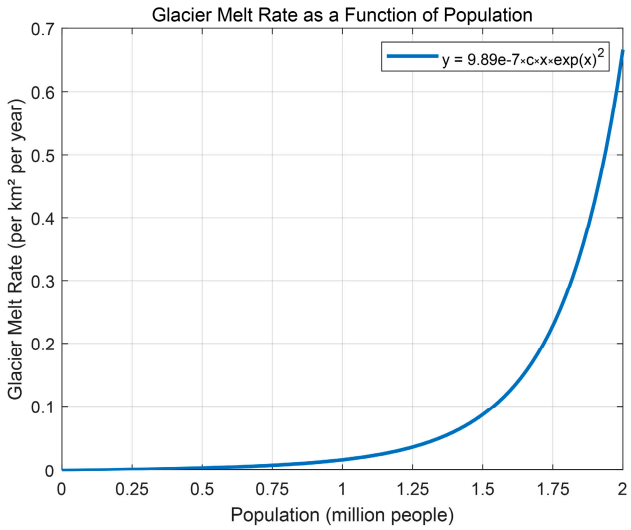


Figure 4. Glacier melting rate versus tourist numbers.

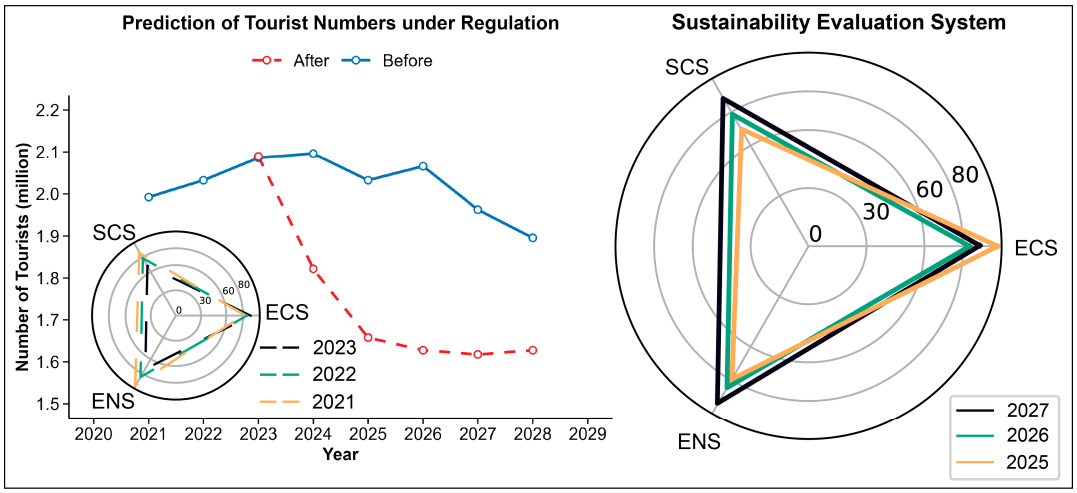


Figure 5. Radar chart of sustainable tourism assessment.

Economically, the revenue function (R(x, r)) scaled near-linearly, rising from USD 200 million at 1 million tourists to USD 400 million at 2 million. Environmentally, the cost function (E(x)) was driven by an exponential increase in the glacier melting rate (L(x)), which, as shown in Figure 4, surged from 100 tons/year at 0.5 million tourists to 2000 tons/year at 1.8 million. Socially, the cost function (S(x)) followed a logistic curve, with the social cost index climbing from 0.15 to 0.75 as tourist numbers increased from 0.5 million to 2 million, driven by rising infrastructure pressure and declining resident satisfaction (from a score of 85 to 50).

The NSGA-II algorithm identified an optimal solution at approximately 1.6 million annual tourists with an 11% tax rate, which yields a revenue of USD 320 million while maintaining environmental and social cost indices at 0.48 and 0.43, respectively. This optimal state contrasts sharply with the current state assessment for 2021–2023 (Figure 5), where recent increases in tourist numbers consistently drove economic scores higher at the expense of steep declines in environmental and social scores. Furthermore, projections based on the optimal solution (Figure 3) show that by 2027, reinvesting tax revenue can decrease environmental and social cost indices to 0.45 and 0.40, respectively, while stabilizing economic revenue at USD 325 million. This demonstrates a convergence toward a balanced and sustainable state, avoiding the boom–bust cycle of a no-intervention scenario.

4. Discussion

4.1. Optimal Scenario and Feedback Mechanism

The sustainable tourism model for Juneau predicts that maintaining annual tourist numbers at approximately 1.6 million, coupled with an 11% tax rate increase, achieves a Pareto-optimal balance across the economic, social, and environmental dimensions. Figure 5 illustrates the projected trends for tourist numbers and dimensional scores from 2024 to 2027 under this optimal solution. Economic revenue is forecasted to stabilize at USD 325 million by 2027. The environmental cost index is projected to decrease from 0.48 in 2023 to 0.45 by 2027, while the social cost index improves from 0.43 to 0.40. These projections incorporate a feedback mechanism where the additional USD 5 million in annual tax revenue is fully reinvested.

The feedback model allocates 60% of the additional revenue (USD 3 million/year) to environmental protection and 40% (USD 2 million/year) to social infrastructure. Environmental investments, such as carbon offset programs and glacier conservation measures, are projected to reduce emissions to 100,000 tibs/year and slow the Mendenhall Glacier's melting rate from 600 tons/year to 550 tons/year, an 8.3% reduction. Social investments focus on infrastructure upgrades, which cumulatively reduce infrastructure pressure from 40% to 35% of capacity and increase resident satisfaction from 68 to 72.

4.2. Comparative Analysis and Projections

To demonstrate the model's utility, a comparison with a no-intervention baseline was conducted (Figure 4). In the baseline scenario (no intervention), tourist numbers would likely rise to 1.9 million by 2025 before cumulative impacts (e.g., accelerated glacier retreat and infrastructure overload) reduce Juneau's appeal, causing a drop to 1.5 million by 2028. This boom–bust cycle, where revenue peaks at USD 380 million and then falls to USD 300 million, mirrors unsustainable patterns in destinations like Venice. In contrast, the intervention scenario stabilizes tourist numbers at 1.6 million before allowing for gradual growth to 1.65 million by 2028 as the destination's carrying capacity improves. This steady approach grows economic revenue to USD 330 million by 2028 while keeping environmental and social cost indices below 0.45 and 0.40, respectively, extending the glacier's tourism viability.

Furthermore, long-term projections (2030–2035) were simulated. With continued feedback, environmental costs could decline to 0.40 by 2030, assuming global emission reductions align with Paris Agreement targets [28]. Social costs may plateau at 0.38 as infrastructure upgrades reach saturation unless new technologies are adopted. Economic revenue could exceed USD 350 million by 2035 if low-carbon attractions boost expenditure. However, climate uncertainties, such as accelerated glacier retreat [29], pose significant risks to these forecasts, highlighting the need for adaptive management.

4.3. Sensitivity and Interdependency Analysis

Sensitivity analysis was conducted to evaluate the interdependencies between key input variables. Figure 6 presents a heatmap visualizing the correlation strengths. The heatmap reveals several critical relationships. Economic revenue (R) exhibited a strong positive correlation with tourist numbers (x) and tax rate (r). Tourist numbers were strongly correlated with infrastructure pressure and social cost (S). Infrastructure pressure itself was strongly correlated with social cost, as capacity overload directly impacted housing availability and traffic congestion [30,31]. Tourist numbers also showed a strong positive correlation with carbon emissions, which, in turn, were strongly negatively correlated with the air quality index. The glacier melting rate exhibited moderate correlations with tourist

numbers and carbon emissions, suggesting indirect effects via regional warming from cruise ship emissions [32].

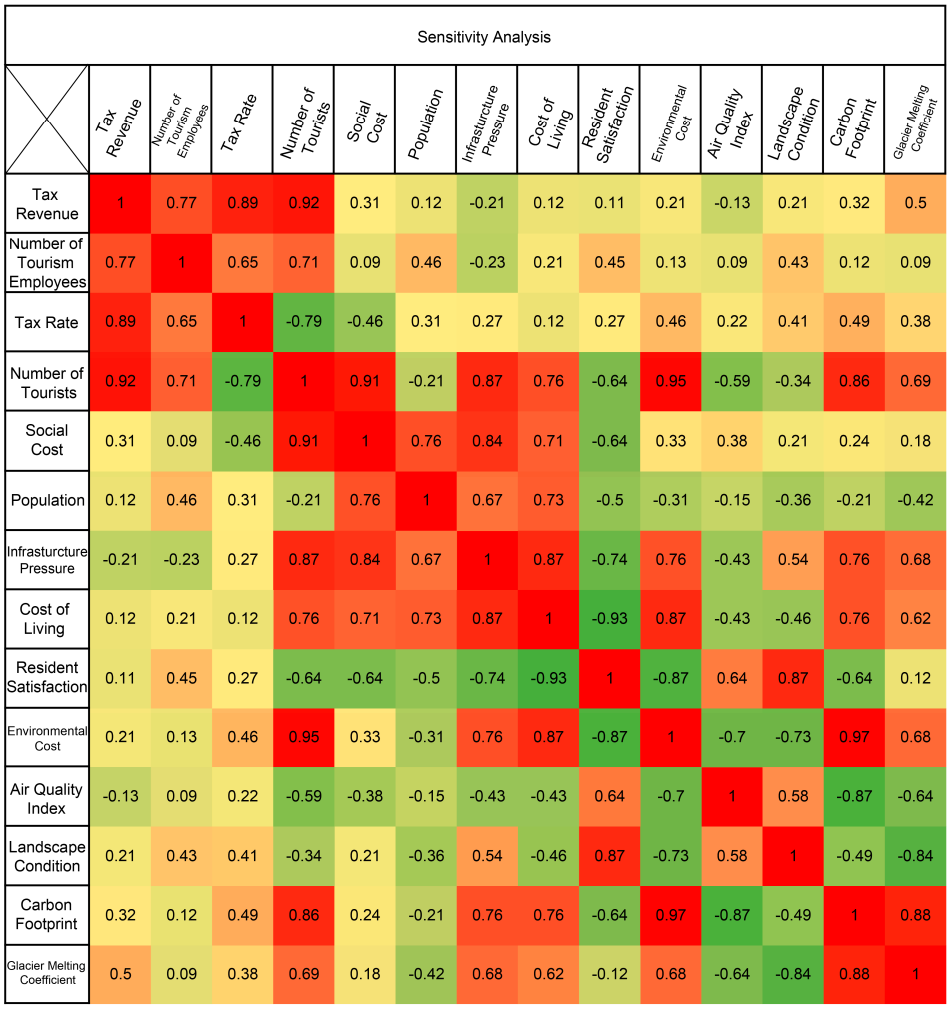


Figure 6. Heatmap of sensitivity analysis. Deeper red indicates stronger positive correlation, and deeper green indicates stronger negative correlation.

To test the model's robustness, input variables were perturbed. A ±10% perturbation of tourist numbers confirmed that 1.6 million is a stable threshold. Increasing tourists to 1.76 million disproportionately increased environmental and social costs (by 25% and 20%, respectively), while reducing them to 1.44 million yielded only marginal cost reductions. Tax rate variations (±2%) altered revenue but had minimal impact on costs (Δindex < 0.02), indicating revenue's insensitivity to small tax changes [33,34]. The analysis also explored interaction effects, showing that combining infrastructure investment and carbon offset funding yielded synergistic cost reductions of 12%. However, glacier melting rates remained resistant to local interventions, highlighting the limits of local action without regional climate policy.

These findings are consistent with overtourism dynamics [17] and underscore the need for low-carbon strategies [28]. The moderate glacier correlation reflects the influence of global climate change, necessitating integration with broader frameworks [29,35].

4.4. Cross-Destination Application: Barcelona

To test the model's adaptability, it was applied to Barcelona, a major urban destination facing distinct overtourism challenges centered on social strain rather than environmental degradation [36]. Key differences include severe housing cost inflation (a 68% rise

from 2014 to 2024) and resident dissatisfaction from noise and congestion [37]. The model was, therefore, recalibrated by modifying the indicator system: social indicators were prioritized, focusing on population density, housing prices, and noise levels, while environmental indicators shifted to air quality, water pollution (WQI), and waste generation. The core multi-objective function and NSGA-II optimization method remained the same [38].

The model predicts an optimal annual tourist threshold of 10 million for Barcelona, a 16.7% reduction from 12 million tourists. Without intervention, numbers could hit 13 million by 2025, escalating the social cost index from 0.60 to 0.70. Under the optimized scenario, a 12% tax rate increase generates EUR 240 million annually for reinvestment (70% social; 30% environmental). By 2027, this stabilizes tourist numbers at 10 million, reducing the social cost index to 0.48 (a 20% improvement) and the environmental cost index to 0.40. Economic revenue dips initially but recovers to EUR 2.5 billion by 2027 as infrastructure enhancements boost visitor satisfaction (Figure 7).

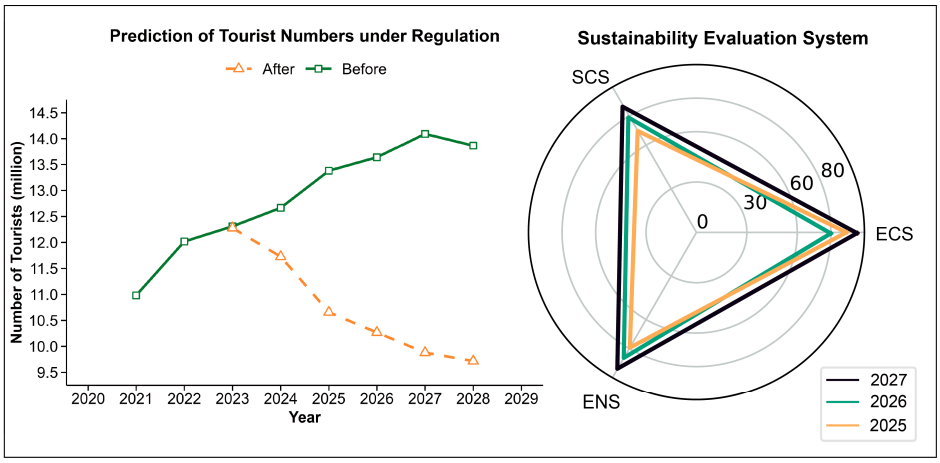


Figure 7. Predictive results for Barcelona's sustainable tourism model.

The model recalibration for Barcelona involved adjusting indicators and weights to reflect urban priorities [39,40]. The social dimension's weight (0.35 total) surpassed that of the environmental and economic dimensions, with housing price stability (0.14) becoming a key indicator. This contrasts with Juneau's environment-focused model (primary weight: 0.30266), which prioritized air quality (0.1374) and glacier health (0.0940).

This cross-destination analysis highlights the framework's flexibility. Juneau's model prioritizes deep environmental protection (e.g., an 8.3% reduction in glacier melting), while Barcelona's model focuses on significant social cost reductions (20%) to address acute urban overtourism, consistent with established destination typologies [41]. Yet, global trends, such as rising sea levels threatening coastal attractions, pose risks, emphasizing the need for regional cooperation with nearby destinations to distribute tourist flows [17,42].

4.5. Model Evaluation and Limitations

4.5.1. Strengths and Weaknesses

The model's primary strength lies in its holistic, multi-dimensional integration, avoiding the oversimplification of single-metric assessments [43]. Its hybrid AHP–entropy weighting method balances expert judgment with objective data, and its flexibility allows for recalibration to diverse destinations like Juneau and Barcelona. Subjective expertise (AHP's consistency ratio: 0.021) can thus be combined with objective data variability. For instance, air quality's high weight (0.1374) reflects its statistical variance in Juneau's 2018–2023 dataset, while expert inputs prioritize glacier protection. This methodology aligns with advanced tourism frameworks, ensuring robust decision-making [37].

However, the model has two main weaknesses. First, its accuracy is highly dependent on data quality. For instance, glacier melting rates may underestimate non-linear climate impacts, and cultural heritage in Barcelona was estimated, not precisely measured, reducing confidence in long-term forecasts [29]. Second, the AHP component introduces subjectivity in weight assignment. Expert panels' varying emphasis on economic versus social goals can influence the model's outputs, highlighting a need for standardized criteria [44].

4.5.2. Limitations and Opportunities for Enhancement

This study is subject to several limitations that also present opportunities for enhancement. The reliance on annual average data masks significant seasonal variations; a dynamic model using real-time data could refine predictions and allow for adaptive policies like monthly visitor caps [45]. Second, the indicator granularity could be improved. Subdividing costs to include metrics like soil erosion, noise pollution, or cultural disruption would offer a more precise picture. Finally, leveraging big data and machine learning could reduce subjectivity in weighting and improve predictive power through iterative analysis of historical data and sentiment analysis [46].

5. Conclusions

This study successfully developed and validated a multi-dimensional sustainable tourism model, providing a quantitative framework for Juneau, Alaska, to navigate the complex trade-offs between economic development and socio-ecological preservation. The model's primary contribution is its predictive ability to identify an optimal balance point for tourism management. For Juneau, our findings demonstrate that optimizing tourist numbers at 1.6 million annually, a modest 5.9% reduction from 2023's 1.7 million, sustains a robust USD 325 million in revenue. This is achieved via an 11% tax increase that generates USD 5 million for reinvestment. This feedback mechanism is critical: allocating 60% (USD 3 million) to environmental protection slows glacier melting to 550 tons/year and lowers the environmental cost index to 0.45 by 2027, while the remaining 40% (USD 2 million) enhances infrastructure, reducing social costs to 0.40 and raising resident satisfaction to a strong score of 76/100.

To validate the framework's adaptability, it was recalibrated and applied to Barcelona, a destination with fundamentally different, socially driven tourism pressures. The model proved highly adaptable, recommending a cap of 10 million (16.7% below 2023's 12 million). This intervention is projected to cut social costs by a significant 20% (index: 0.48) through EUR 168 million in housing and transport upgrades. Crucially, economic viability is maintained, with revenue recovering to EUR 2.5 billion by 2027. Barcelona's social focus (weight: 0.35) contrasts with Juneau's environmental priority (0.30266), confirming the model's flexibility and utility across diverse urban and natural contexts.

Methodologically, the rigor of this model is underpinned by the AHP–entropy weighting method (consistency ratio: 0.021), which balances objective data with expert input to prioritize critical indicators like air quality (weight: 0.1374) in Juneau. However, despite these strengths, the framework's precision is subject to limitations. Data gaps (e.g., underestimated glacier loss) and the inherent subjectivity of the AHP suggest that future iterations could be enhanced via real-time analytics and dynamic modeling.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/su17167344/s1; File S1: Primary_Data; File S2: Analysis_and_Supplement.

Author Contributions: Conceptualization, S.L.; Writing—original draft, J.P.; Writing—review & editing, H.Y., Z.W. and B.P. All authors have read and agreed to the published version of the manuscript.

Funding: Natural Science Foundation of China (32171844) and Innovation Capacity Building Project of Beijing Academy of Agriculture and Forestry Sciences (KJCX20230306).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflict of interest.

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